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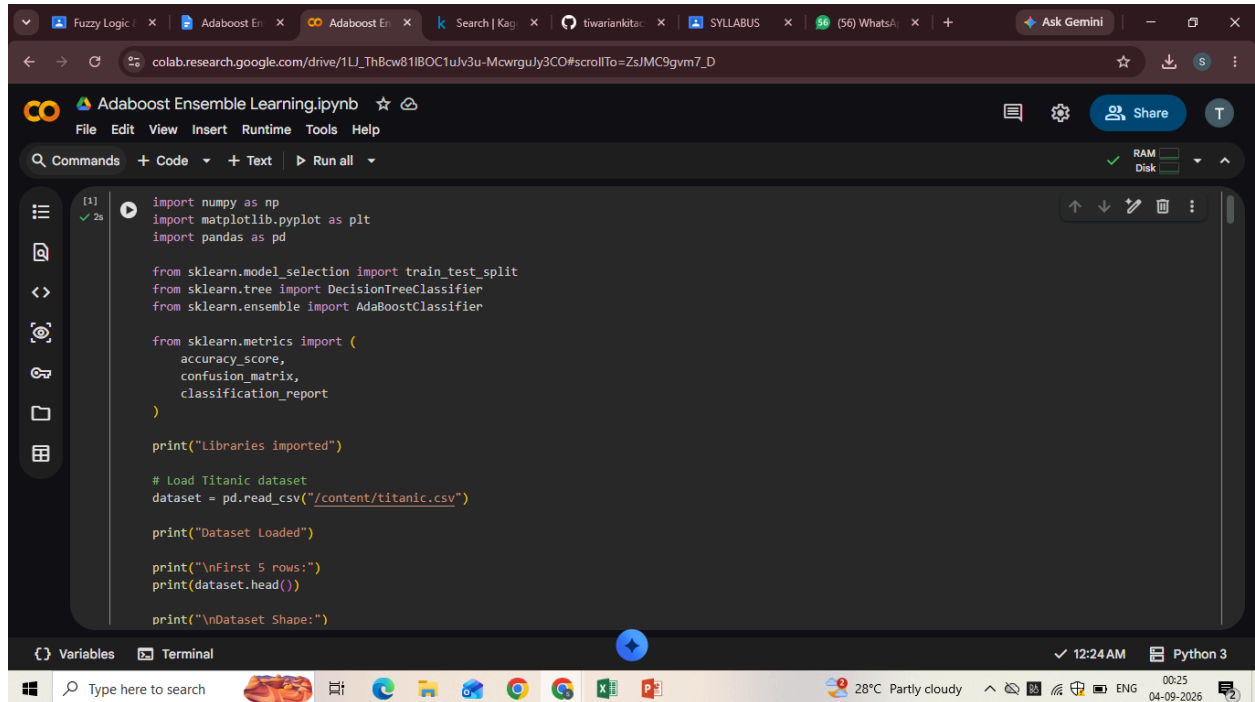
AIM: Adaboost Ensemble Learning

Implement the Adaboost algorithm to create an ensemble of weak classifiers.

Train the ensemble model on a given dataset and evaluate its performance.

Compare the results with individual weak classifiers.

INPUT:



```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import AdaBoostClassifier

from sklearn.metrics import (
    accuracy_score,
    confusion_matrix,
    classification_report
)

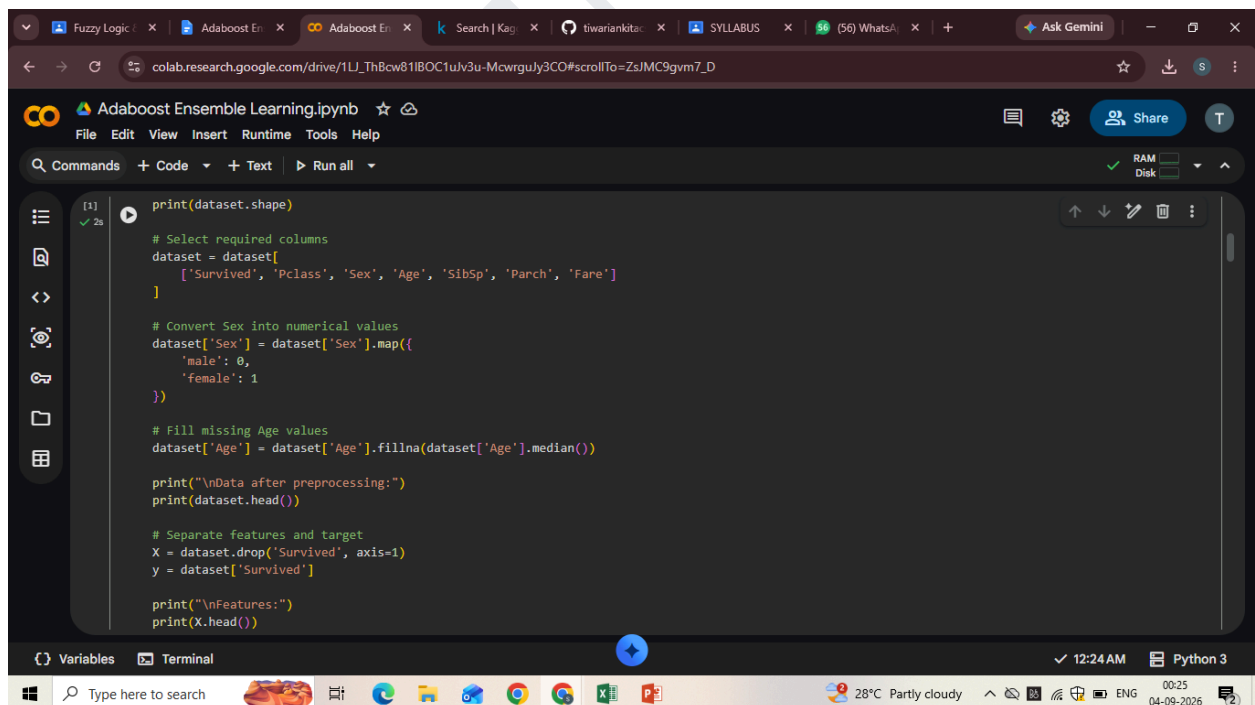
print("Libraries imported")

# Load Titanic dataset
dataset = pd.read_csv("/content/titanic.csv")

print("Dataset Loaded")

print("\nFirst 5 rows:")
print(dataset.head())

print("\nDataset Shape:")
```



```
print(dataset.shape)

# Select required columns
dataset = dataset[
    ['Survived', 'Pclass', 'Sex', 'Age', 'SibSp', 'Parch', 'Fare']
]

# Convert Sex into numerical values
dataset['Sex'] = dataset['Sex'].map({
    'male': 0,
    'female': 1
})

# Fill missing Age values
dataset['Age'] = dataset['Age'].fillna(dataset['Age'].median())

print("\nData after preprocessing:")
print(dataset.head())

# Separate features and target
X = dataset.drop('Survived', axis=1)
y = dataset['Survived']

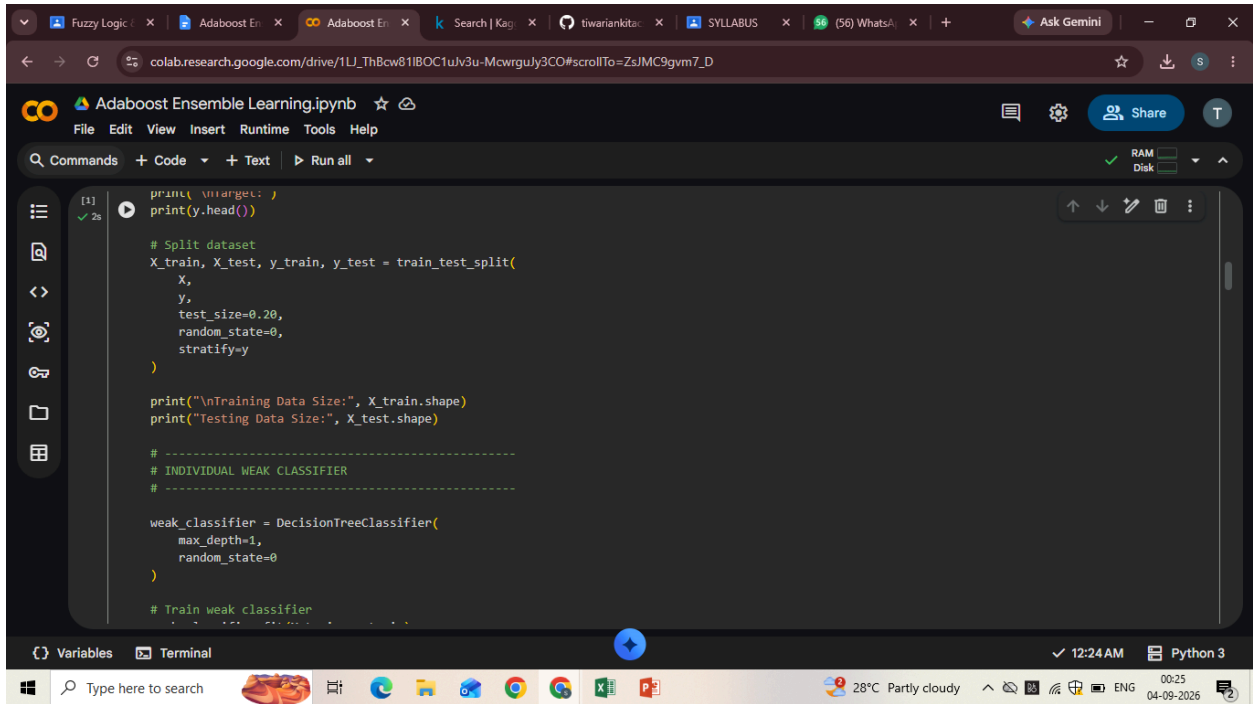
print("\nFeatures:")
print(X.head())
```

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```
print(' \n\n target: ', y)
print(y.head())

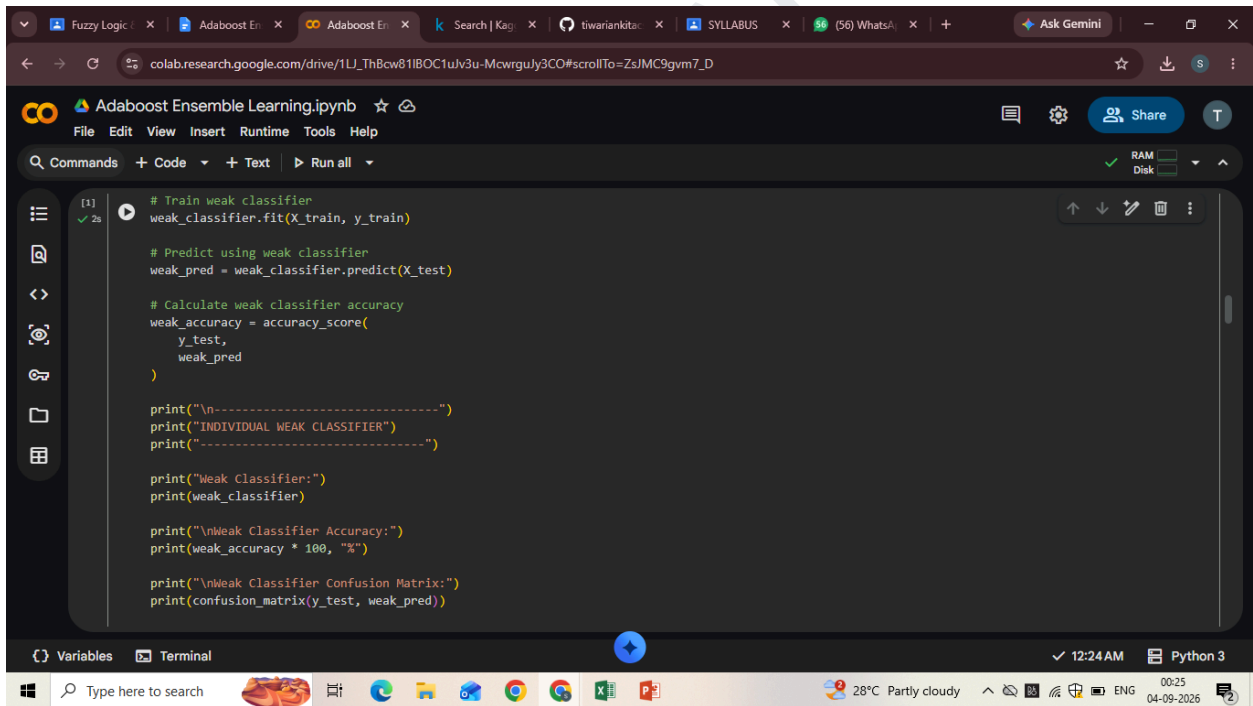
# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=0,
    stratify=y
)

print('\n\n Training Data Size:', X_train.shape)
print('\n\n Testing Data Size:', X_test.shape)

# -----
# INDIVIDUAL WEAK CLASSIFIER
# -----

weak_classifier = DecisionTreeClassifier(
    max_depth=1,
    random_state=0
)

# Train weak classifier
```



```
# Train weak classifier
weak_classifier.fit(X_train, y_train)

# Predict using weak classifier
weak_pred = weak_classifier.predict(X_test)

# Calculate weak classifier accuracy
weak_accuracy = accuracy_score(
    y_test,
    weak_pred
)

print('\n\n -----')
print('\n\n INDIVIDUAL WEAK CLASSIFIER')
print('\n\n -----')

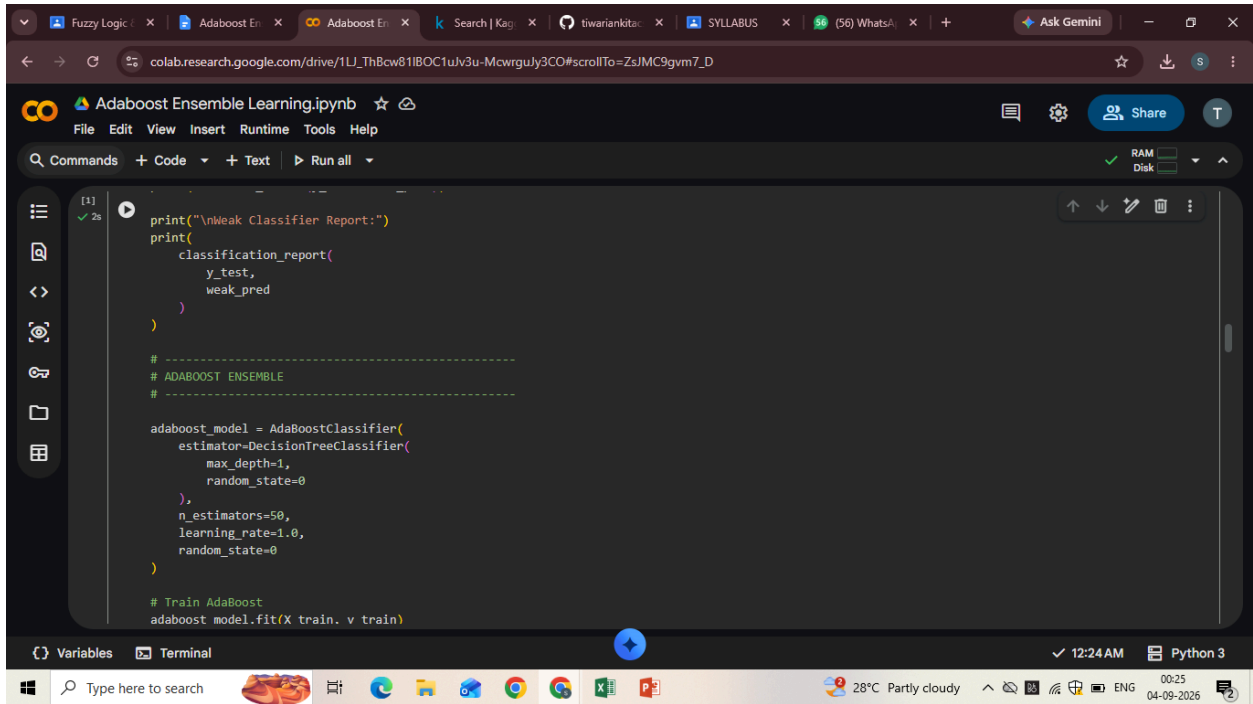
print('\n\n Weak Classifier:')
print(weak_classifier)

print('\n\n Weak Classifier Accuracy:')
print(weak_accuracy * 100, '%')

print('\n\n Weak Classifier Confusion Matrix:')
print(confusion_matrix(y_test, weak_pred))
```

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The screenshot shows a Jupyter Notebook interface with the following code:

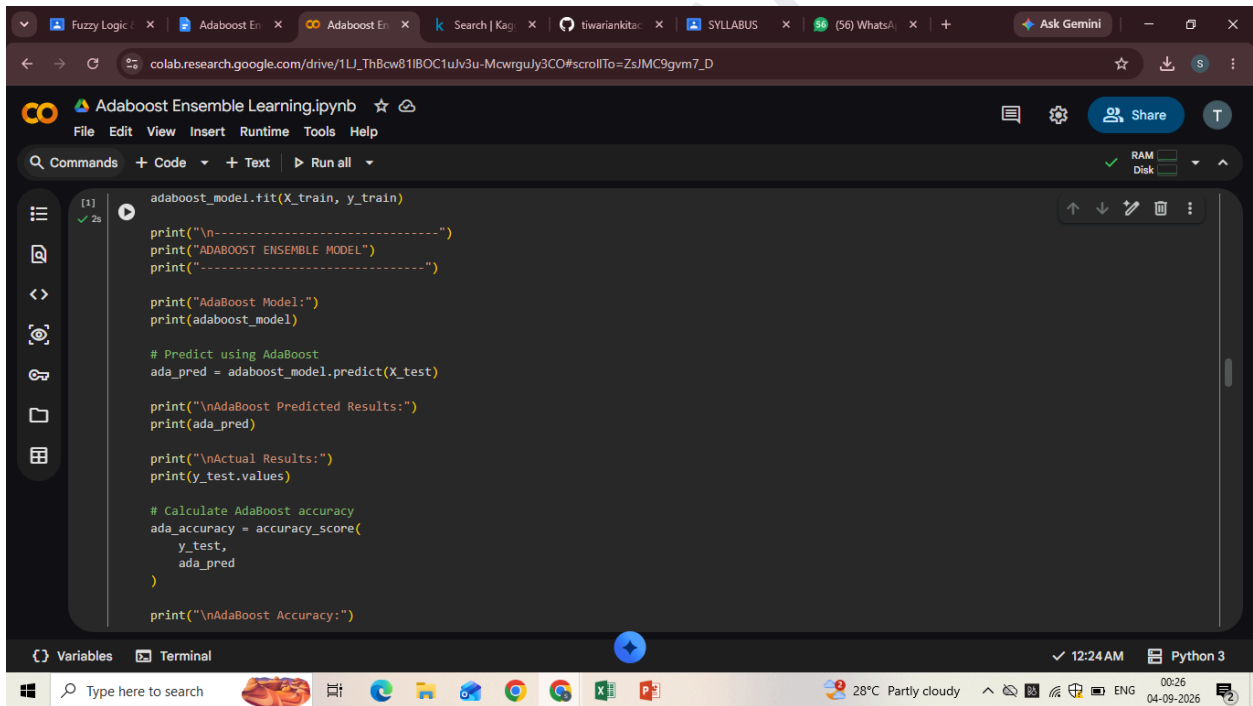
```
[1] ✓ 2s
print("\nWeak Classifier Report:")
print(
    classification_report(
        y_test,
        weak_pred
    )
)

# -----
# ADABOOST ENSEMBLE
# -----

adaboost_model = AdaBoostClassifier(
    estimator=DecisionTreeClassifier(
        max_depth=1,
        random_state=0
    ),
    n_estimators=50,
    learning_rate=1.0,
    random_state=0
)

# Train AdaBoost
adaboost_model.fit(X_train, y_train)
```

The interface includes a top bar with browser tabs, a file explorer on the left, and a bottom status bar showing the time as 12:24 AM and Python 3.



The screenshot shows the same Jupyter Notebook with the following code:

```
[1] ✓ 2s
adaboost_model.fit(X_train, y_train)

print("\n-----")
print("ADABOOST ENSEMBLE MODEL")
print("-----")

print("AdaBoost Model:")
print(adaboost_model)

# Predict using AdaBoost
ada_pred = adaboost_model.predict(X_test)

print("\nAdaBoost Predicted Results:")
print(ada_pred)

print("\nActual Results:")
print(y_test.values)

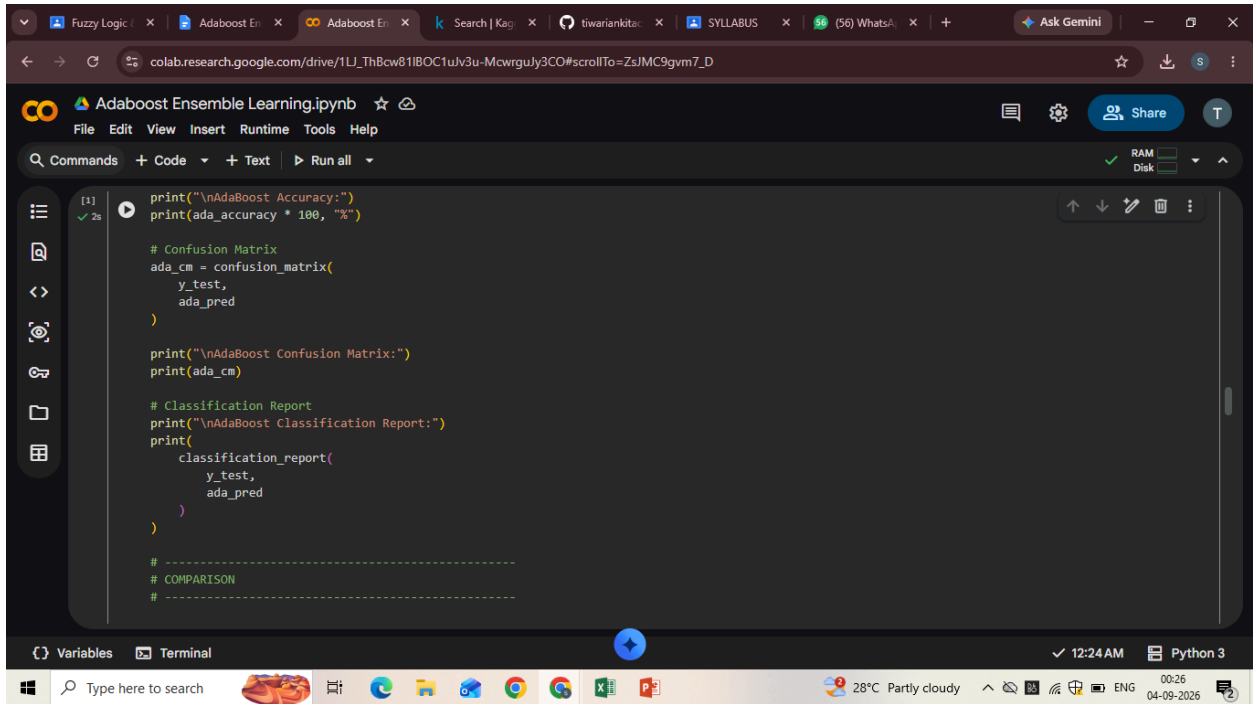
# Calculate AdaBoost accuracy
ada_accuracy = accuracy_score(
    y_test,
    ada_pred
)

print("\nAdaBoost Accuracy:")
```

The interface is identical to the first screenshot, showing the same top bar, file explorer, and bottom status bar.

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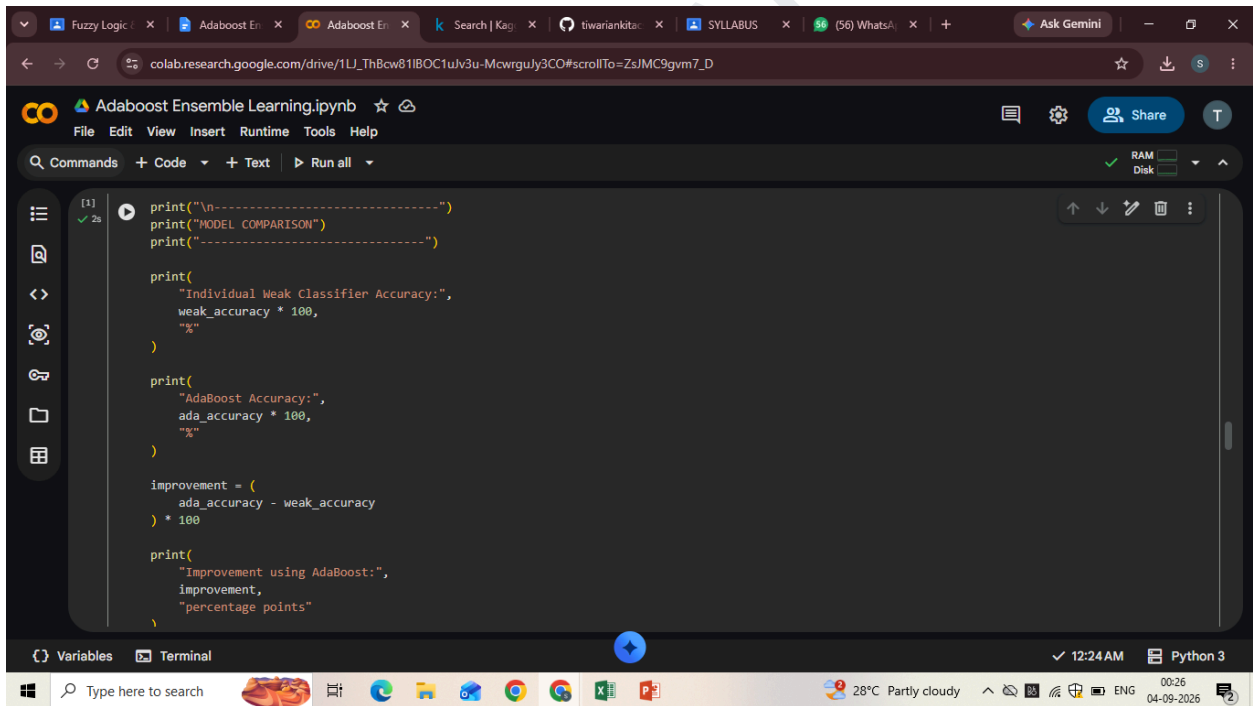
```
[1] ✓ 2s
print("\nAdaBoost Accuracy:")
print(ada_accuracy * 100, "%")

# Confusion Matrix
ada_cm = confusion_matrix(
    y_test,
    ada_pred
)

print("\nAdaBoost Confusion Matrix:")
print(ada_cm)

# Classification Report
print("\nAdaBoost Classification Report:")
print(
    classification_report(
        y_test,
        ada_pred
    )
)

# -----
# COMPARISON
# -----
```



```
[1] ✓ 2s
print("\n-----")
print("MODEL COMPARISON")
print("\n-----")

print(
    "Individual Weak Classifier Accuracy:",
    weak_accuracy * 100,
    "%"
)

print(
    "AdaBoost Accuracy:",
    ada_accuracy * 100,
    "%"
)

improvement = (
    ada_accuracy - weak_accuracy
) * 100

print(
    "Improvement using AdaBoost:",
    improvement,
    "percentage points"
)
```

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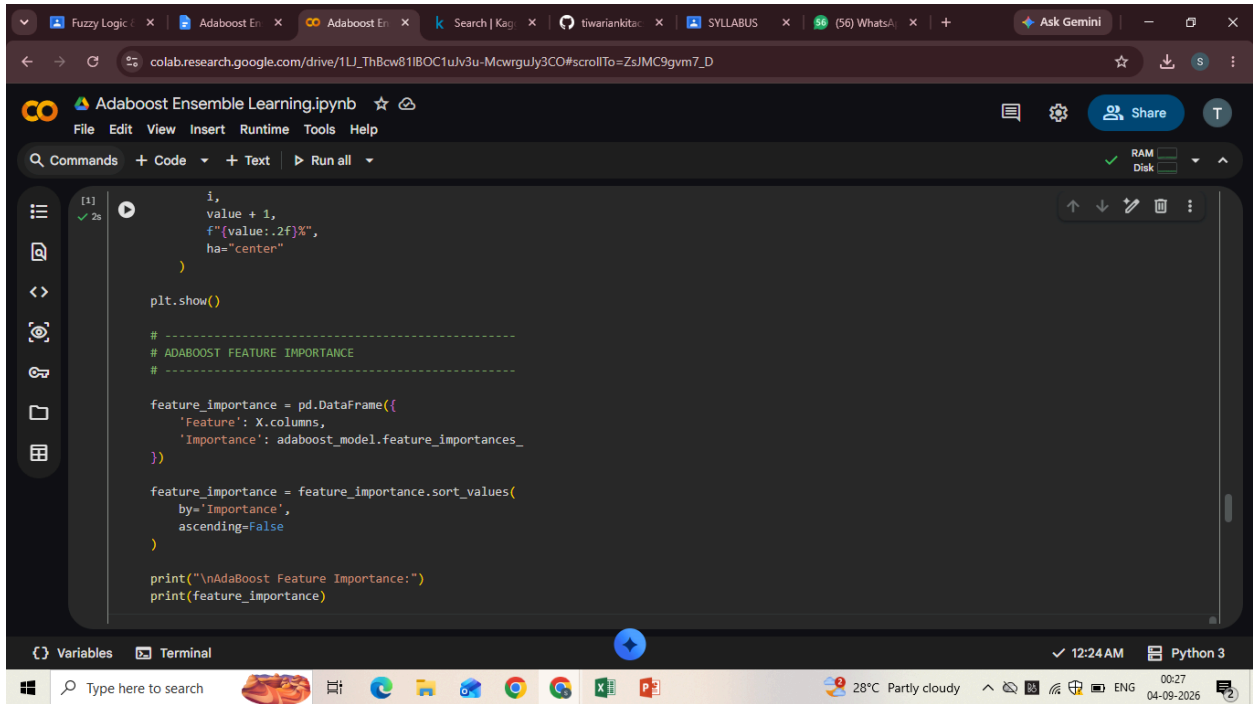
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```
# -----  
# ACCURACY COMPARISON GRAPH  
# -----  
  
models = [  
    "Weak Classifier",  
    "AdaBoost"  
]  
  
accuracies = [  
    weak_accuracy * 100,  
    ada_accuracy * 100  
]  
  
plt.figure(figsize=(8, 5))  
  
plt.bar(  
    models,  
    accuracies  
)  
  
plt.xlabel("Model")  
  
plt.ylabel("Accuracy (%)" )
```

```
plt.ylabel("Accuracy (%)")  
  
plt.title("Weak Classifier vs AdaBoost")  
  
plt.ylim(0, 100)  
  
for i, value in enumerate(accuracies):  
    plt.text(  
        i,  
        value + 1,  
        f"{value:.2f}%",  
        ha="center"  
    )  
  
plt.show()  
  
# -----  
# ADABOOST FEATURE IMPORTANCE  
# -----  
  
feature_importance = pd.DataFrame({  
    'Feature': X.columns,  
    'Importance': adaboost_model.feature_importances_  
})
```

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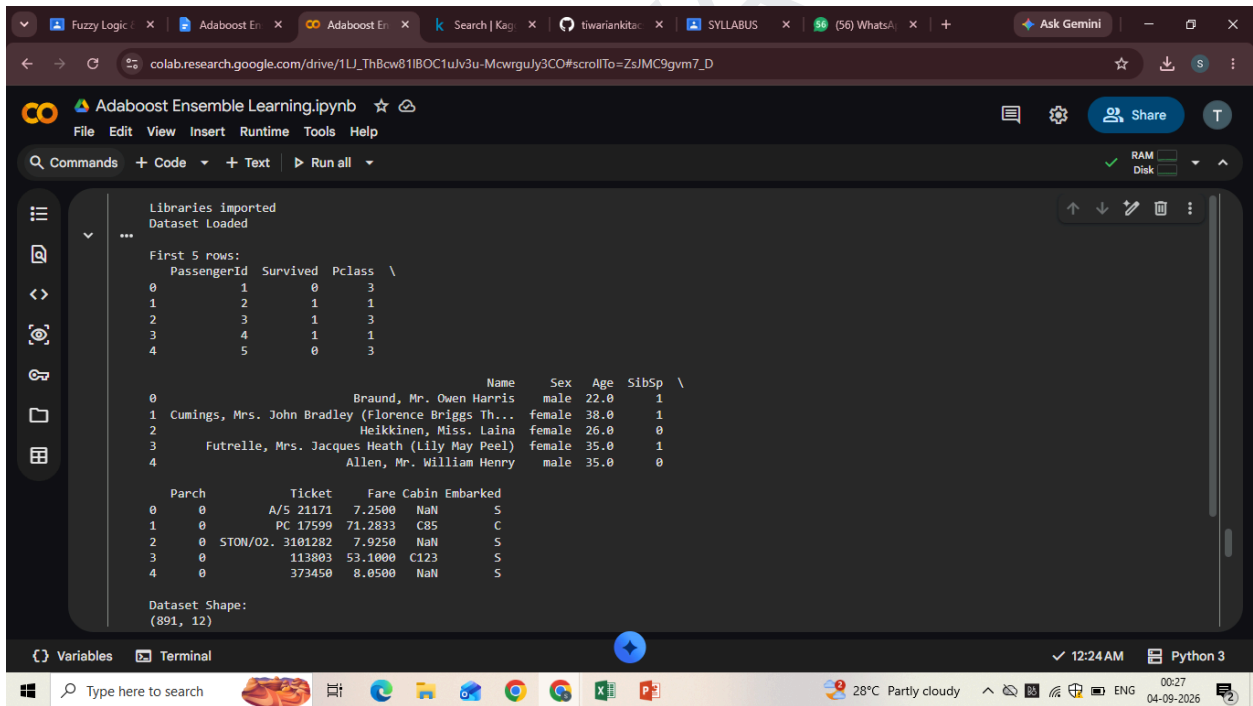


The screenshot shows a Jupyter Notebook interface with the following code in the first cell:

```
[1] ✓ 2s  
    i,  
    value + 1,  
    f"{value:.2f}%",  
    ha="center"  
)  
  
plt.show()  
  
# -----  
# ADABOOST FEATURE IMPORTANCE  
# -----  
  
feature_importance = pd.DataFrame({  
    'Feature': X.columns,  
    'Importance': adaboost_model.feature_importances_  
})  
  
feature_importance = feature_importance.sort_values(  
    by='Importance',  
    ascending=False  
)  
  
print("\nAdaboost Feature Importance:")  
print(feature_importance)
```

The bottom of the notebook shows the status bar with "Variables", "Terminal", "12:24 AM", and "Python 3".

OUTPUT:



The screenshot shows the output of the Jupyter Notebook code. The output is as follows:

```
Libraries imported  
Dataset Loaded  
...  
First 5 rows:  
PassengerId  Survived  Pclass  \  
0            1         0        3  
1            2         1        1  
2            3         1        3  
3            4         1        1  
4            5         0        3  
  
0            Braund, Mr. Owen Harris    male  22.0    1  
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0    1  
2            Heikinen, Miss. Laina    female  26.0    0  
3  Futrelle, Mrs. Jacques Heath (Lily May Peel)    female  35.0    1  
4            Allen, Mr. William Henry    male  35.0    0  
  
Parch      Ticket      Fare  Cabin  Embarked  
0            0      A/5 21171    7.2500  NaN      S  
1            0      PC 17599   71.2833   C85      C  
2            0  STON/O2. 3101282    7.9250  NaN      S  
3            0   113803   53.1000  C123      S  
4            0   373450    8.0500  NaN      S  
  
Dataset Shape:  
(891, 12)
```

The bottom of the notebook shows the status bar with "Variables", "Terminal", "12:24 AM", and "Python 3".

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The top screenshot shows the initial dataset shape and preprocessing steps. The dataset shape is (891, 12). The data after preprocessing is shown as a table with columns: Survived, Pclass, Sex, Age, SibSp, Parch, Fare. The target variable is Survived, dtype: int64.

	Survived	Pclass	Sex	Age	SibSp	Parch	Fare
0	0	3	0	22.0	1	0	7.2500
1	1	1	1	38.0	1	0	71.2833
2	1	3	1	26.0	0	0	7.9250
3	1	1	1	35.0	1	0	53.1000
4	0	3	0	35.0	0	0	8.0500

The bottom screenshot shows the training data size (712, 6) and testing data size (179, 6). The individual weak classifier is a DecisionTreeClassifier with max_depth=1 and random_state=0. The weak classifier accuracy is 77.6536312849162 %.

Weak Classifier Confusion Matrix:

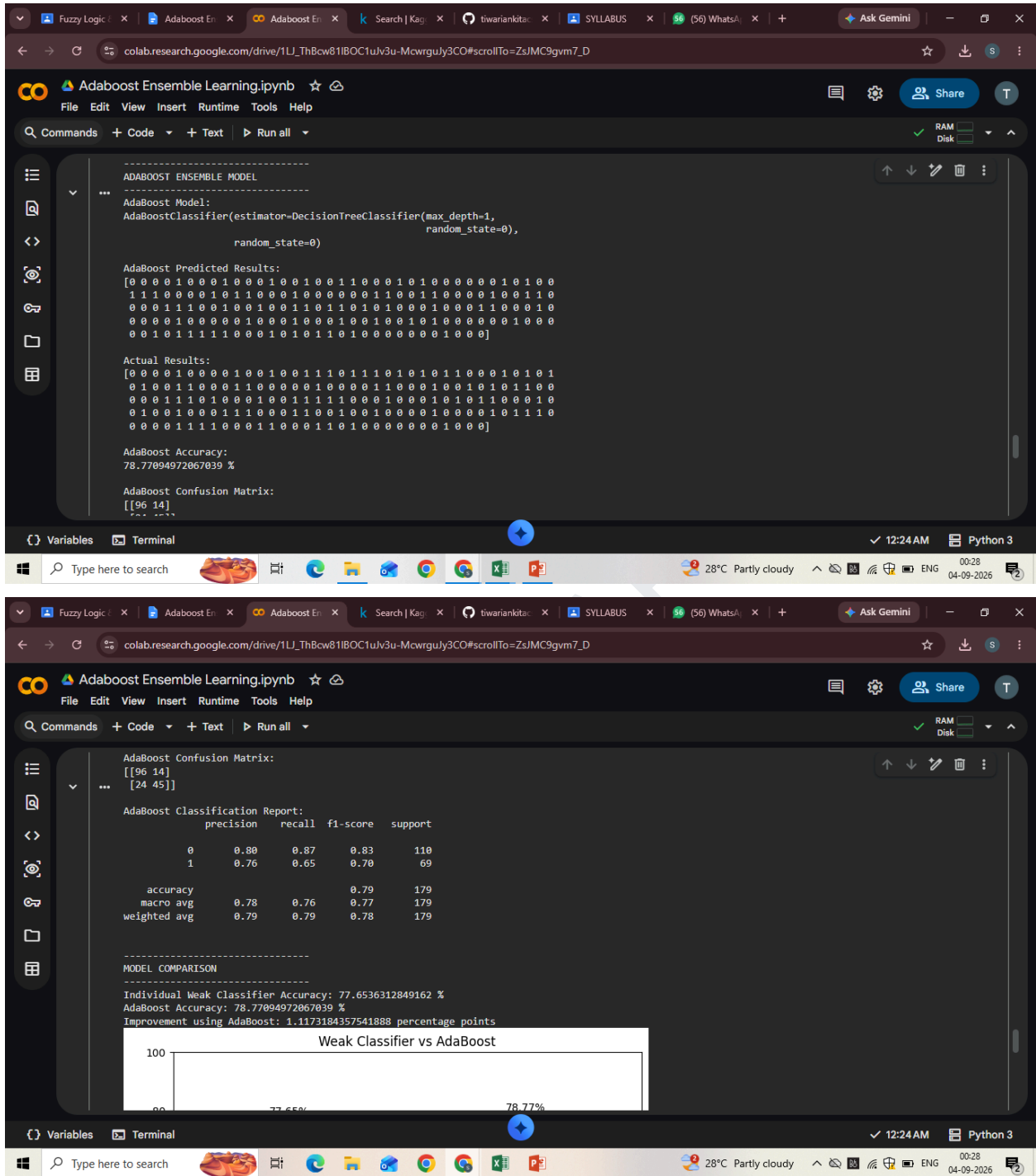
		0	1	
0	193	17		
1	23	46		

Weak Classifier Report:

	precision	recall	f1-score	support
0	0.80	0.85	0.82	110
1	0.73	0.67	0.70	69
accuracy			0.78	179
macro avg	0.77	0.76	0.76	179
weighted avg	0.77	0.78	0.77	179

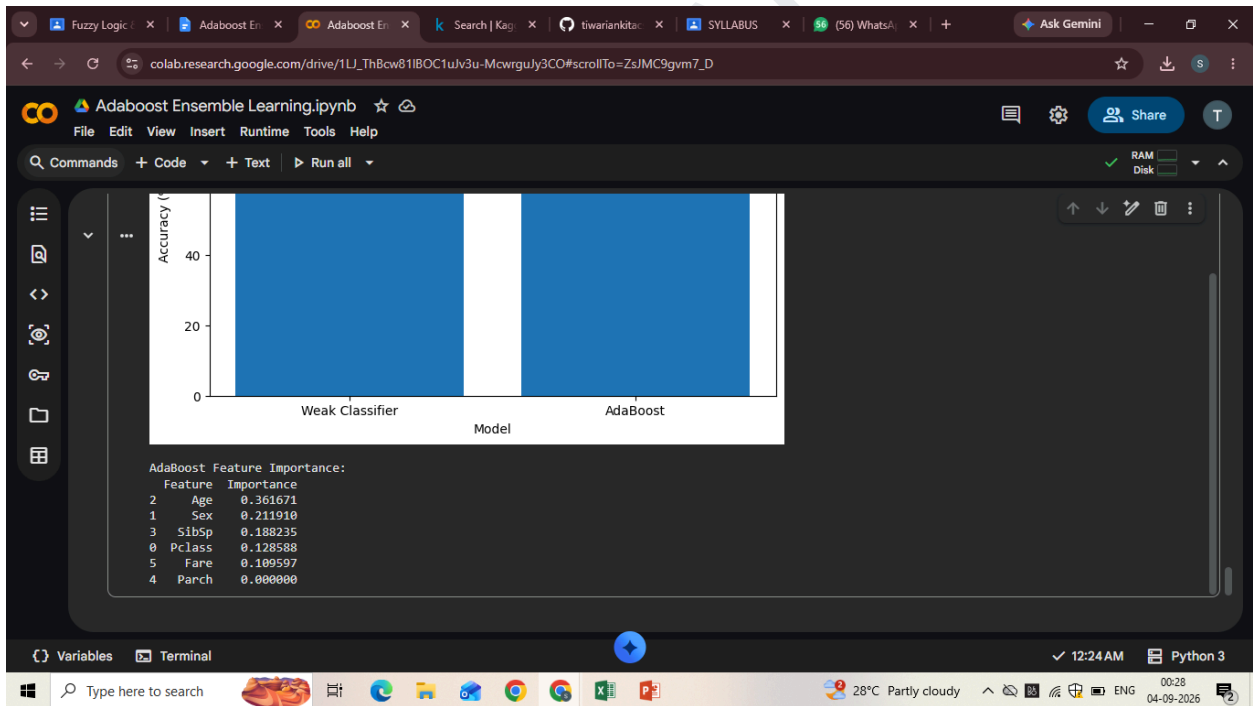
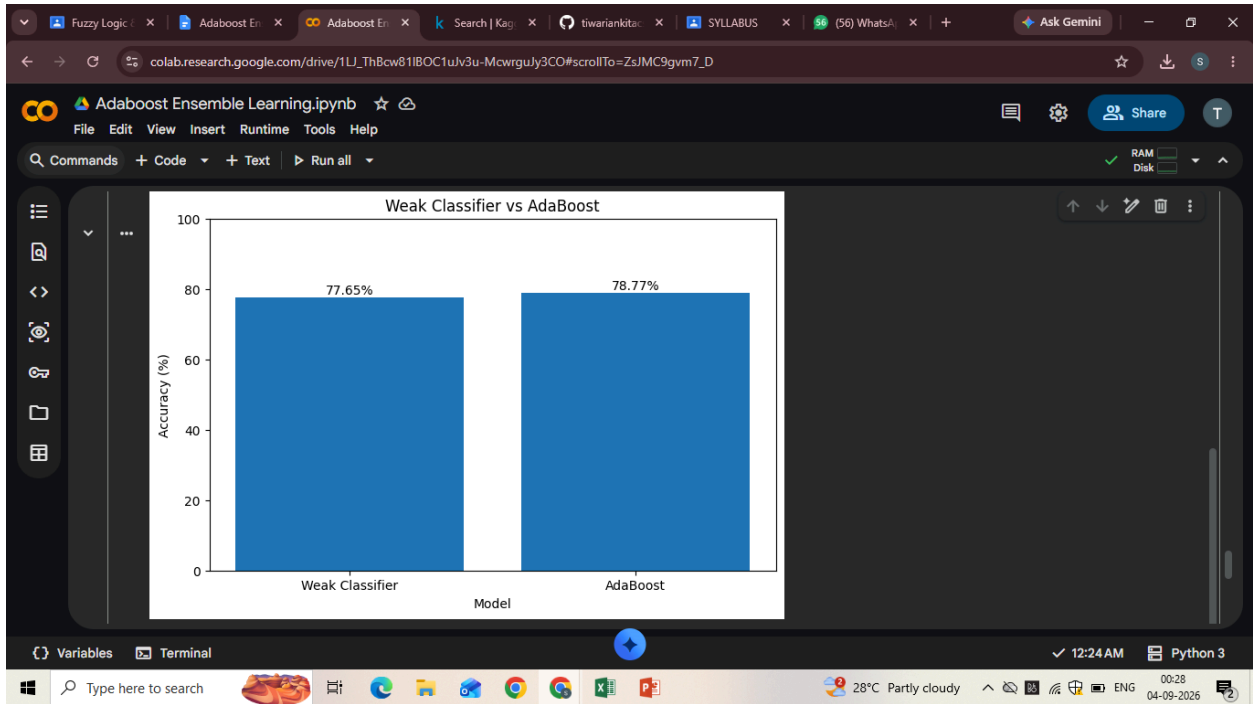
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